Customer Churn Analytics
Business Insights Summary

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1. Overview

This document translates the churn-prediction analysis into business-facing insights. It identifies the leading causes of churn, profiles the customers most at risk, estimates the revenue exposed, and recommends targeted retention actions. All findings are drawn from the Telco Customer Churn dataset and the selected Logistic Regression model.

Headline finding: roughly one in four customers churns (26.5%). Churn is concentrated among new, fiber-optic, month-to-month customers, while long-tenure and contract-locked customers rarely leave. The model flags 428 high-risk customers carrying $33,545.30 in monthly recurring revenue.

2. Top Churn Reasons

The Logistic Regression coefficients reveal which factors push customers toward leaving (positive) versus staying (negative). Magnitude indicates strength of influence.

Factor	Coefficient	Effect on Churn
Fiber-optic internet	**+0.83**	Strongest churn driver; fiber customers leave most
Month-to-month contract	**+0.32**	No lock-in; easy to cancel
High total charges	**+0.49**	Interpret with care (see note)
Streaming movies / TV	**+0.27 / +0.26**	Add-ons linked to higher churn
Electronic-check payment	**+0.12**	Associated with less stable customers
Longer tenure	**−1.16**	Strongest protective factor
Two-year contract	**−0.34**	Lock-in strongly reduces churn
Online security add-on	**−0.13**	Bundled security retains customers
Has dependents	**−0.10**	Household accounts are stickier

Important interpretation note: high total charges shows a positive coefficient, but total charges is closely correlated with tenure (which is strongly protective). In a linear model these two partially offset each other, so high spending should not be read as a direct cause of churn. The cleaner takeaway is that low tenure — being a newer customer — is the dominant risk signal.

3. High-Risk Customer Profile

Combining the strongest churn drivers produces a clear profile of the customer most likely to leave:

- **Tenure:** new or short-tenure (under ~12 months).
- **Internet:** fiber-optic service — by far the largest single risk factor.
- **Contract:** month-to-month, with no lock-in.
- **Payment:** electronic check rather than automatic payment.
- **Add-ons:** lacking online security and tech support; may have streaming add-ons.
- **Charges:** relatively high monthly charges for a short relationship.

By contrast, the most loyal customers are long-tenure, on one- or two-year contracts, using automatic payment, and subscribed to security or support add-ons that increase switching friction.

4. Customer Risk Segmentation

Every customer is assigned a churn probability and placed in one of three risk tiers, each with a recommended action priority:

Risk Tier	Probability	Customers	Action Priority
High	0.66 – 1.00	428	Urgent outreach
Medium	0.33 – 0.66	352	Monitor / nurture
Low	0.00 – 0.33	629	Standard service

The 428 High-risk customers are the priority group: they represent the greatest probability of loss and therefore the highest expected return on retention effort.

5. Revenue Impact

Monthly recurring revenue at risk: $33,545.30

This figure is the combined monthly charges of the 428 customers the model places in the High-risk tier within the test set. It represents recurring revenue that would be lost each month if these customers churned and were not retained.

5.1 How to Read This Number

- It is a monthly figure, not annual — the annualised exposure for this group is roughly twelve times larger.
- It covers the 20% test sample only; scaled across the full customer base, total exposure is substantially higher.
- It is potential exposure, not certain loss: not every high-risk customer will actually churn, and successful retention recovers part of this revenue.

5.2 The Retention Economics

Because the model favours recall, it casts a wide net: of the customers it flags, some would have stayed anyway. This is deliberate. Reaching a few extra customers with a retention offer is far cheaper than silently losing a customer who was never contacted. The 81 churners the model still misses (from the test set) represent the residual gap to close with richer data over time.

6. Recommended Actions

6.1 Immediate

- Prioritise the 428 High-risk customers for proactive retention outreach — targeted offers, loyalty incentives, or service check-ins.
- Encourage migration from month-to-month to one- or two-year contracts using discounts, given the strong protective effect of contract lock-in.
- Investigate the fiber-optic experience: as the single largest churn driver, service quality, pricing, or onboarding for fiber customers warrants review.

6.2 Medium Term

- Promote automatic-payment enrolment and bundle online-security / tech-support add-ons, both linked to lower churn.
- Focus early-lifecycle engagement on new customers, since low tenure is the dominant risk signal.
- Tune the model's decision threshold to match the retention budget — a higher threshold reduces false alarms, a lower one catches more churners.

6.3 Data & Modelling

- Begin collecting usage and support-interaction data to enable the behavioural features the original brief envisioned but the current dataset lacks.
- Re-train the model periodically as new churn outcomes accumulate.

7. Summary

Churn is concentrated among new, fiber-optic, month-to-month customers paying by electronic check, while long-tenure contract-bound customers rarely leave. The Logistic Regression model identifies about four out of five churners and isolates 428 high-risk customers carrying $33,545.30 in monthly recurring revenue. Directing retention effort at this group — alongside contract-conversion and fiber-experience improvements — offers the clearest path to reducing churn and protecting revenue.